Question One

ZCAS University has recently established a cutting-edge laboratory dedicated to advancing research facilities in the field of Software Engineering. Beyond catering to the needs of research scholars and professors, the laboratory has become a hub for a considerable number of employees involved in various projects initiated by the institution. With the university's continuous growth in size and capacity, there is a compelling proposal to develop a Library Information System (LIS) aimed at enhancing the experience of both students and employees. The LIS is designed to streamline the process of borrowing and returning books, allowing users the convenience of managing these transactions from their desks or chambers. Additionally, the system empowers members to extend the borrowing period, contingent on the absence of prior bookings for the specific book. For the library staff, this system serves as a valuable tool for efficiently handling day-to-day book transactions. The librarian, possessing administrative privileges and complete system control, can effortlessly input new records for recently acquired books or remove records when a book is no longer available. While the system offers non-members the freedom to browse and search for books online, the issuance and return of books remain exclusive to valid users (members) of the LIS. This initiative represents a strategic integration of technology to foster a more accessible and efficient library experience within the dynamic environment of ZCAS University.

You are required to model the above system using Deployment Diagram. **[25 Marks]**

Question Two

Design and implement a class hierarchy to represent different types of shapes, including circles, rectangles, and squares. Each shape should have methods to calculate its area and perimeter. Additionally, implement an interface called "Drawable" that specifies a method to draw the shape on a graphics context. All shapes should implement the Drawable interface. **[25 Marks]**

Requirements:

1. Create a base class called "Shape" with abstract methods for calculating area and perimeter.
2. Create subclasses for Circle, Rectangle, and Square, each inheriting from the Shape class.
3. Implement the abstract methods from Shape in each subclass, providing the appropriate calculations for each shape type.
4. Implement the Drawable interface in each subclass, providing the code to draw the corresponding shape on a graphics context.
5. Create a test class to instantiate objects of each shape type and call their area, perimeter, and draw methods.
6. Demonstrate method overloading by creating overloaded methods for calculating the area of rectangles based on different combinations of parameters (length, width, or both).
7. Utilize inheritance to ensure that all shape objects can be drawn using the Drawable interface's draw() method.

Question Three

The university has decided to develop its own WatCard system, rather than rely on software from a vendor. The initial release of the software allows a WatCard holder to use his or her card to gain access to secure areas on campus (e.g., their residence after hours, computer rooms). He or she can also use the card as a debit card, meaning that he/she can deposit money into an associated account, and can purchase items in certain locations by using the card to withdraw money from the account. In the beginning, students will only be able to use their WatCards to pay various university-related fees: parking fees (at parking permit dispensers), library fees, equipment rental fees at the Physical Activity Centre, fees for official transcripts, etc. Also, in the beginning, only online deposits will be supported. The system keeps a Usage record of every use of the WatCard.

Use your Java programming knowledge to implement this system by using the following OOP concepts:

i. Inheritance **[5 Marks]**
ii. Interfaces **[5 Marks]**
iii. Method overloading **[5 Marks]**
iv. Polymorphism **[5 Marks]**
v. Method overriding **[5 Marks]**

Question Four

i. Create a class that has instance variables of String and Integers which can respectively store a student's first name and age. In the class, create a constructor that assigns non default values to these variables. **[10 Marks]**

ii. Modify the StudentClass to make its constructor behave like a default constructor. **[5 Marks]**

iii. Create a time class that has 3 private instance variables hour, minute and second. Your class should have overloaded constructors that can be used to manipulate these instance variables. **[10 Marks]**

Question Five

i. Create an executable class to an instantiate object of StudentClass2 and allow users to enter their first name and age. Display the values of the object's instance variables

before and after user's input.

Marks]

ii. Create a class that has a String instance variable and an int instance variable, in the class, create a constructor that has parameters that uses these instance variables as its data. **[10 Marks]**

Question Six

Create a class called Invoice that a hardware store might use to represent an invoice for an item sold at the store. An Invoice should include four pieces of information as instance variables a part number (type String), a part description (type String), a quantity of the item being purchased (type int) and a price per item (double). Your class should have a constructor that initializes the four instance variables. Provide a set and a get method for each instance variable. In addition, provide a method named getInvoiceAmount that calculates the invoice amount (i.e., multiplies the quantity by the price per item), then returns the amount as a double value. If the quantity is not positive, it should be set to 0. If the price per item is not positive, it should be set to 0.0. Write a test application named InvoiceTest that demonstrates class Invoice's capabilities.

[25 Marks]